

Chengyu (Harry) Yu

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[LinkedIn](#)

EDUCATION

Carnegie Mellon University

M.S. in Computer Science

- GPA: 3.72/4.00

Pittsburgh, PA

Aug 2025 – Dec 2026

Wesleyan University

B.A. in Computer Science & Film Studies

- GPA: 4.0/4.0 | Phi Beta Kappa | High Honors in Computer Science

- Wesleyan Freeman Asian Scholar (full merit-based scholarship; only recipient from China in Class of 2025)

Middletown, CT

Sep 2021 – Jun 2025

SKILLS

- **Programming:** Python, TypeScript/JavaScript, C, C++, Standard ML/OCaml, Shell, Java
- **ML & GenAI:** PyTorch, Diffusion models, DiT, ICL, RLHF, DPO, Transformers, RAG, TinyBERT, LoRA fine-tuning
- **Frameworks & Tools:** React, React-Native, Node.js, Redux, Expo, Numpy, TensorFlow.js, Git, Jest/PyTest, GitHub Actions
- **Databases / Cloud:** MySQL, BigQuery, Firebase

EXPERIENCES

Generative AI Machine Learning Engineer Intern | ByteDance (Incoming)

Starting May 2026

- Scoped to develop Multimodal LLM Agents for POI (Point-of-Interest) review automation, integrating vision-language models (VLM) with tool-use and RAG pipelines to verify merchant images, addresses, and category metadata at scale

Computer Vision / Generative AI Machine Learning Engineer Intern | Hithink RoyalFlush

Jun 2024 – Aug 2024

- Independently trained and deployed 5+ style LoRA models from the ground up on H800 GPU server. Automated end-to-end model training pipeline for data collection, labeling and weight adjusting through (YOLOv5, WD-14, GPT-4o) — delivered models to >1M users.
- Built internal diffusion-model GUI in ComfyUI, improved system stability by adjusting ComfyUI's queue API and workflow integrity check to block malicious queue prompts, raising system uptime from 92% to 99.8%.
- Conducted in-depth research in Video Generation model frameworks on Diffusion Transformers (DiTs) architectures and spatial-temporal VAE. Published in company's internal research report for 5000+ employees.

Lead Researcher | Privacy Pioneer

Jun 2024 – Apr 2026

- Designed and trained 10+ transformer-based TinyBERT models for real-time privacy analysis of network traffic, achieving 96% F1 score in detecting HTTP snippets that track user location.
- Improved extension's classification accuracy by 40%+ via standardizing escape sequences and optimizing tokenizer input pipelines for HTTP message preprocessing.
- Developed JS-based Selenium crawler to conduct automatic web privacy detection on 10,000+ sites; deployed geo-distributed analysis across 10+ countries via Google Cloud Virtual Machines.

Student Researcher | Cross-platform Election Advertising Transparency Initiative, NSF

Dec 2022 – Sep 2024

- Collaborated on the design and implementation of a standardized political ad data platform funded by the National Science Foundation; utilizing ML classifiers to standardize nearly all cross-platform Internet political ads for federal elections (2M+ rows). Co-authored the research paper *Comparable 2022 General Election Advertising Datasets from Meta and Google*, published in *Nature's Scientific Data*.
- Conducted code functionality tests across 20+ repositories and 200+ scripts, integrated ML classifiers (SVM, Random Forest) and ABSA into ETL pipeline. Implemented standardized repository documentation; designed frontend architecture and created instructional video tutorials.

Lead Front-End Developer | OurCampusWes — Wesleyan University

Sep 2023 – Jun 2024

- Led a complete bottom-up redesign and redevelopment of OurCampusWes, the university's exclusive social media platform, delivering core features including WesMarketplace (student item exchange) and WesRide (student transportation), alongside professor evaluations, course tracking, and campus event updates.
- Added asynchronous storage & Firebase Cloud Messaging for real-time course-seat alerts; tripled daily active usage of the course module.
- Switched linear search to Fuse.js fuzzy search with seat-based sorting, slashing average query latency from 800 ms to 60 ms.

PUBLICATIONS

- **(Under Review)** Ning, J., Li, X., & Yu, C. (Equal contribution) (2026). *Revision or Re-Solving? Decomposing Second-Pass Gains in Multi-LLM Pipelines*. *arXiv preprint*, arXiv:2604.01029. <https://arxiv.org/abs/2604.01029>
- Zhang, M., Cakmak, F., Neumann, M., Zimmeck, S., Oleinikov, P., Yao, J., Yu, H., Jacewicz, A., Tassone, I., Floyd, B., Baum, L., Franz, M. M., Ridout, T. N., & Fowler, E. F. (2025). *Comparable 2022 General Election Advertising Datasets from Meta and Google*. *Scientific Data (Nature Portfolio)*, 12, 968. <https://doi.org/10.1038/s41597-025-05228-w>
- Yu, H., Yin, P., & Zimmeck, S. (2026). *Global Web, Local Privacy? An International Review of Web Tracking*. *Pragmatic Cybersecurity*, 1(1). <https://www.sciltp.com/journals/pc/articles/2603003347>

HONORS AND AWARDS

- Dean's List (7 Semesters), Wesleyan University 2021 – 2025
- Michael Rice Prize in Computer Science (one of five winners at Wesleyan University) May 2025
- 2022 Johnston Prize in Physics (one of three winners at Wesleyan University) Apr 2022